Band-1 Hadamard Order 10164 – Classical Paley-I (q=10163); IN SPK via Paley-I; Gap 2052 Still OPEN

H_10164 Paley-I PASS; Paley-II SKIP (q=5081 prime ===1 mod 4 eligible; prefer Paley-I); Kronecker FAIL/SKIP (4*2541; 2541===1 mod 4 NOT Hadamard-admissible); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-23

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-23_Band1_Hadamard_10160_PaleyI`; `2026-09-23_Band1_Hadamard_10152_PaleyI`; `2026-09-23_Band1_Hadamard_10140_PaleyI`; `2026-09-23_Band1_Hadamard_10112_PaleyI`; `2026-09-23_Band1_Hadamard_10104_PaleyI`; `2026-09-23_Band1_Hadamard_10100_PaleyI`; `2026-09-23_Band1_Hadamard_10068_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_10164_paleyI/`.
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Honesty stamp (non-negotiable).
This memo claims **finite construction of order 10164 only (verified max|HHˆT-nI|=0), via classical Paley-I q=10163 (prime ===3 mod 4). 10164 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 10164: Paley-I **PASS** ($q = 10163$ prime $\equiv 3 \pmod 4$); Paley-II **SKIP** ($q = 5081$ prime $\equiv 1 \pmod 4$ eligible; prefer Paley-I); Kronecker **FAIL/SKIP** (note $10164 = 4 \cdot 2541$; $2541 \equiv 1 \pmod 4$ NOT Hadamard-admissible). **10164** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **10160+10152+10140+10112+10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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Abstract

Finite construction of order **10164**.

- H_{10164} = Paley-I ($q = 10163$) – classical SPK
- Paley-II for 10164: **SKIP** ($q = 5081$ prime $\equiv 1 \pmod 4$ eligible; prefer Paley-I)

- Kronecker for 10164: **FAIL/SKIP** ($4 \cdot 2541$; $2541 \equiv 1 \pmod 4$ NOT Hadamard-admissible)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_10164.py`

Results. overall_pass=True; units 211/211; row blocks 80/80 (block size 128); wall 858.2618s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 10164 = q + 1$ with $q = 10163$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 5081$ prime $\equiv 1 \pmod 4$ — arithmetically eligible; SKIP construction this wave (prefer Paley-I).
- Kronecker: $10164 = 4 \cdot 2541$; $2541 \equiv 1 \pmod 4$ NOT Hadamard-admissible — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_10164 Paley-I	Paley-I q=10163	0	839.3538
H_10164 Paley-II	SKIP (q=5081 prime ===1 mod 4 eligible; prefer Paley-I)	–	–
H_10164 Kronecker	FAIL/SKIP (4*2541; 2541===1 mod 4 NOT Hadamard-admissible)	–	–

SPK closure note

Seeds include prior SPK through **10160** lineage (**10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**;

9324; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10164** (+ prior **10160+10152+10140+10112+10104+10100+10068+10040+10008+9968+9932+9924+9908+9888+9**
Next constructive gap **2052** (OPEN).

Quantity	Value
10164 in SPK	**True** (Paley-I q=10163)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=10211->n=10212 Paley-I DIRECT q=10211 OR q=10223->n=10224 Paley-I DIRECT q=10223 OR q=10243->n=10244 Paley-I DIRECT q=10243 OR n=2052 standard SPK constructive gap – method-attack / lit OR Track B / Object 1 meter await